# Kartik Patel

Brampton, Ontario, Canada | +358 40 3230116 • kartikpat25@gmail.com
| https://www.linkedin.com/in/kpatel250905/ | https://github.com/KartikPat250905 | https://kartikworks.netlify.app/

## ABOUT

Second-year Computer Science student with hands-on experience building AI-driven backend systems at a startup and shipping a published Android app with 200+ active users. Combines academic coursework with real-world software development experience across backend, ML, and full-stack projects.

## TECHNICAL SKILLS

- **Programming :** Python, C++, C, C#, MicroPython, Rust, R.
- **Frontend Technologies :** React, JavaScript, TypeScript, HTML, CSS.
- **ML Libraries:** Tensorflow, NumPy, Pandas, Matplotlib.
- **Other Technologies :** Git, CI/CD, Linux, Bash, Firebase, Excalidraw, Unity, MariaDB.

## EDUCATION

**B.Sc. in Computer Science CO-OP, CGPA (4.06/4.33)** Jan 2026 - Present
Toronto Metropolitan University, Toronto, Canada

**Bachelor of Engineering in ICT, CGPA (5.0/5.0)** Aug 2024 - May 2025
Metropolia University of Applied Sciences, Helsinki, Finland

**School 42 peer-learning software engineering school** Feb 2024 - Jul 2025
Hive, Helsinki, Finland

## WORK EXPERIENCE

**Geek Squad Agent** Aug 2026 - Present
Best Buy, Brampton, Canada

- Diagnosed and resolved hardware, software, and connectivity issues for customers, applying a CS background to deliver device setup, repair, and troubleshooting support while maintaining strong customer satisfaction.

**Junior AI Developer** Jun 2025 - Feb 2026
Cydora, Helsinki, Finland
Technologies used: **Python, SQL, REST APIs, CI/CD, Git, AI/ML libraries.**

- Built and maintained AI-driven backend systems automating workflows for 4 client accounts in production.
- Developed ML pipeline components integrated into backend services handling 2,000+ requests.
- Improved system reliability by introducing structured testing and CI/CD practices, reducing manual processing time by an estimated ~25%.

**AI Intern** Mar 2025 - Jun 2025
Cydora, Helsinki, Finland
Technologies used: **Python, React.js, Web Development tools, AI Agents, AI/ML libraries.**

- Supported integration of AI features into 2 client-facing applications, contributing to both backend and frontend workflows.
- Prototyped and evaluated 3 AI tools/agents to improve development efficiency, shortening a key internal workflow by ~15%.
- Assisted in testing and debugging components across the stack, helping reduce reported bugs by ~20% ahead of client delivery.

## PROJECTS

**Loot Radar | Kotlin Multiplatform, Firebase (Auth, Firestore) |** https://github.com/KartikPat250905/loot-radar

- Built and published Loot Radar (Kotlin Multiplatform), a game giveaway tracker with 200+ active users, using a diff-then-filter pipeline for precise, exactly-once notifications.
- Implemented Firebase Auth and Firestore for secure user accounts and real-time cross-device sync of filter preferences
- Designed a filter-matching engine that evaluates each detected giveaway against thousands of per-user platform/type combinations at scale.

**Heartbeat Sensor | MicroPython| Group Project |** https://github.com/ktnlvr/cardiotron

- Developed a heartbeat monitoring system using ADC signals on Raspberry Pi Pico for real-time heartbeat detection and analysis.
- Implemented signal filtering and exported processed data to Kubios for advanced heart rate variability (HRV) analysis.

**Fetal HC Segmentation | Python, Tensorflow, R |** https://github.com/KartikPat250905/fetal-hc-segmentation

- Designed and trained a U-Net CNN with a custom BCE-Dice loss for pixel-wise segmentation of fetal skull boundaries in ultrasound imagery.
- Built an end-to-end tf.data preprocessing and training pipeline with checkpointing, early stopping, and learning-rate scheduling.
- Validated predicted measurements, achieving 97% segmentation accuracy (Dice) and 2.65% mean measurement error.

**Minishell | C, Bash | Duo Project |** https://github.com/KartikPat250905/minishell

- Built a simplified Unix shell replicating core bash functionality in C.
- Implemented command parsing, tokenization, and process execution.
- Handled signal management and memory-safe resource cleanup.

## HACKATHON

**Supercell AI Hackathon | Unity, C# | 3rd place | AI/Game Development hackathon**

- Won 3rd place at the Supercell AI Hackathon for creating a prototype where cat-controlled player interacts with context-aware, AI driven character
- Recognized among 50+ global teams for designing a standout real time AI interaction system that blends gameplay narrative, and LLM based character behavior.

**Student Network Challenge Hackathon | 3rd place | Idea hackathon**

- Hackathon organized by Microsoft, Nokia and Elisa and hosted by University of Helsinki.
- Conceptualized an AI-driven solution for sustainability, securing 3rd place among multiple teams.

**Junction Hackathon | Python (ML) |** https://github.com/ktnlvr/junction-2024

- Developed a model that predicts input amounts based on water loss at each production stage to minimize end-product standard deviation.
- Achieved 97% accuracy with the provided dataset.

## LEETCODE

**LeetCode | Data Structures and Algorithms | C++ |** https://leetcode.com/u/KartikPat25094/

- Solved over 230 problems on LeetCode.
- Experienced in advanced algorithms: Backtracking and Divide and Conquer.
- Strong knowledge of Trees, Binary Trees, DFS, and other data structures : Arrays, Linked Lists, Stacks, Tries.